

Supply Chain and Inventory Optimization

Advanced portfolio project

Flag overstock, stockout risk, reorder candidates, sell-through, and value-based ABC classes.

project_id: supply_chain_inventory | template_id: powerbi_kpi_slicer | source_rows: 48 | prepared_rows: 48 | generated_at: 2026-08-11T06:45:08.314608+00:00

Executive summary

Supply Chain and Inventory Optimization analyzed 48 rows across 16 fields. The reusable powerbi_kpi_slicer BI template presents source-backed KPIs, drivers, and detail tables. Total purchases is 5,443 with a transparent synthetic validation fixture or uploaded source.

Headline metrics

Rows analyzed

48

Total purchases

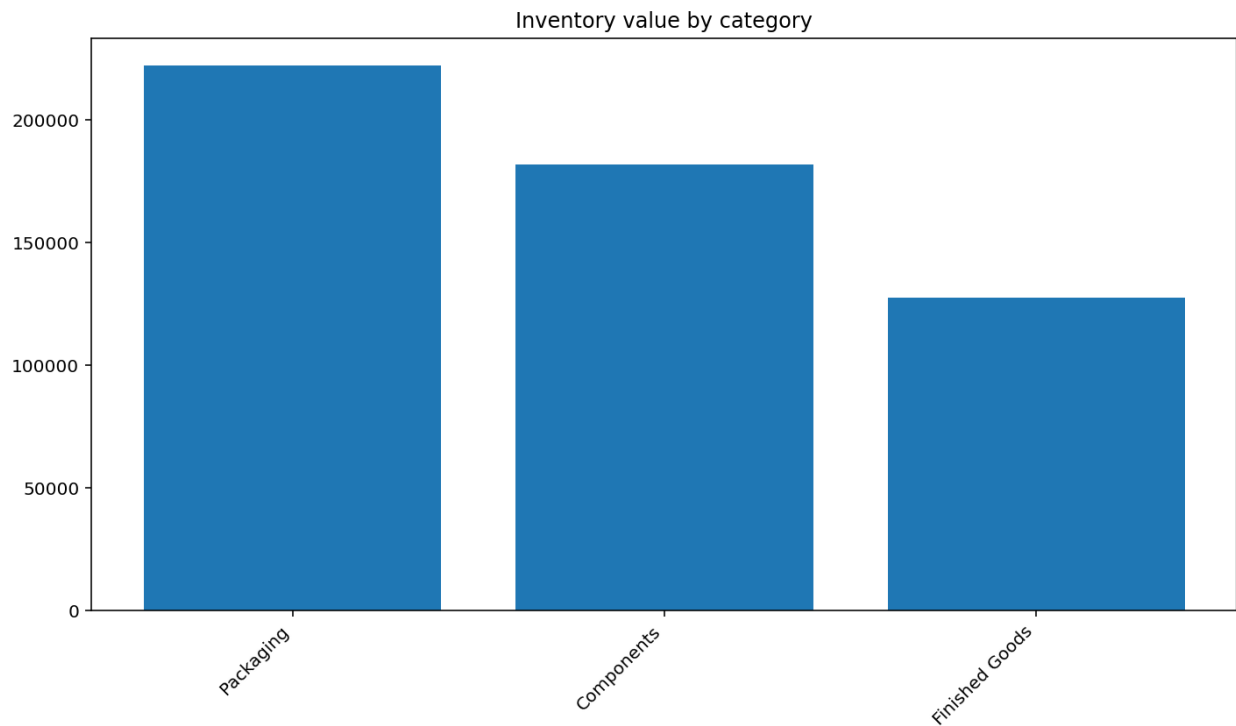
5,443

Average purchases

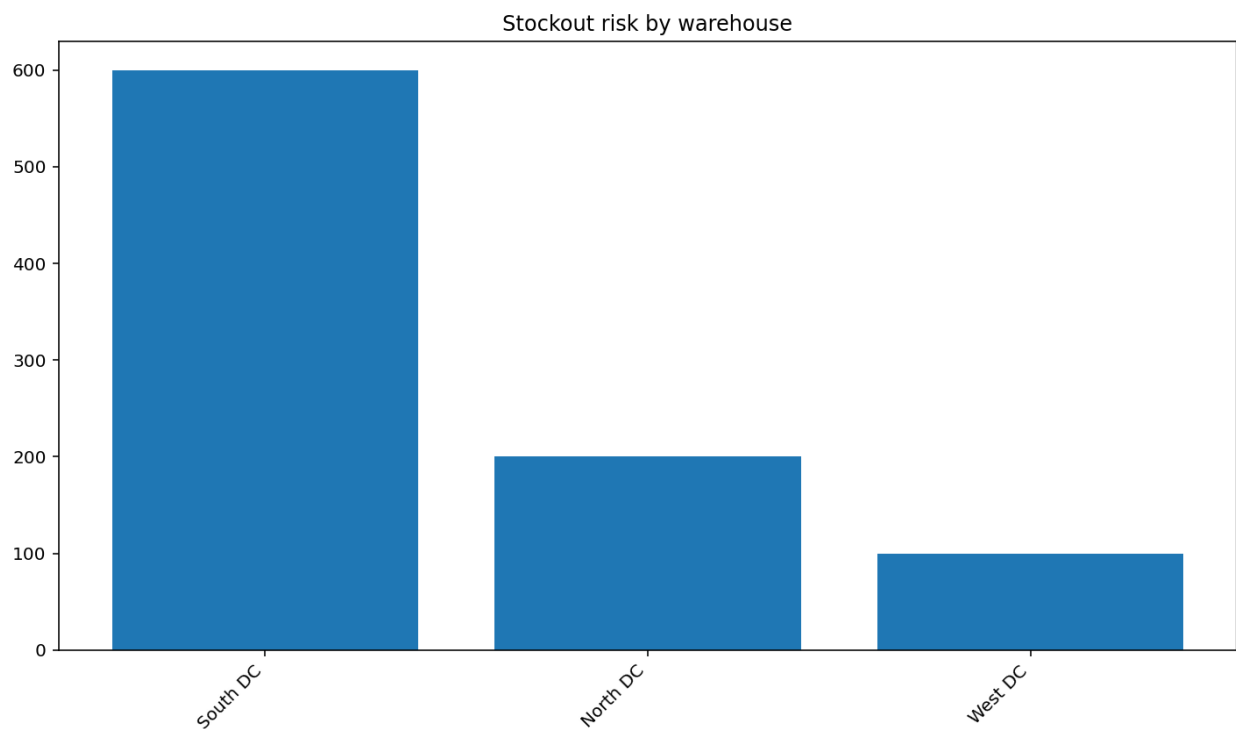
113.40

Drivers and trends

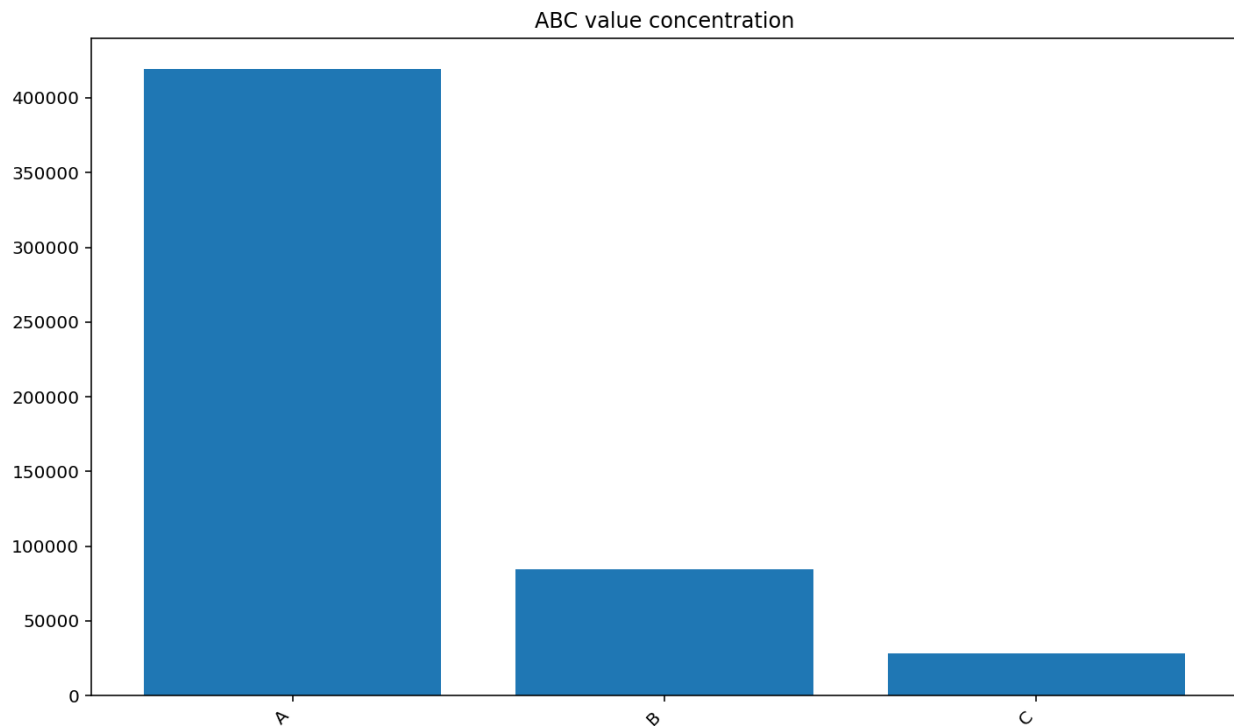
Inventory value by category



Stockout risk by warehouse



ABC value concentration



Detail by category

category	count	sum	mean
Packaging	16	222308.67	13894.291875
Components	16	181974.66	11373.41625
Finished Goods	16	127555.85	7972.240625

Detail by warehouse

warehouse	count	sum	mean
South DC	16	600.0	37.5
North DC	16	200.0	12.5
West DC	16	100.0	6.25

Key findings

abc_class explains substantial variation in inventory_value ($\eta^2=0.727$).; category and warehouse have strong categorical association (Cramér's $V=1.000$).; sku and category have strong categorical association (Cramér's $V=1.000$).; sku and product_name have strong categorical association (Cramér's $V=1.000$).; sku and warehouse have strong categorical association (Cramér's $V=1.000$).; opening_stock and closing_stock are strongly correlated ($r=0.873$).; abc_class explains substantial variation in closing_stock ($\eta^2=0.516$).; date is nearly unique per row (100.0%).

Caveats and assumptions

Fixtures are synthetic validation data. Uploaded datasets remain the source of truth; field aliases are recorded in the build metadata.